

AMENDMENT NO. 1 MARCH 2026
TO
IS 548 (PART 3/SEC 1) : 2021 METHODS OF SAMPLING AND TEST FOR OILS
AND FATS
PART 3 ADVANCED INSTRUMENTAL METHODS
SECTION 1 DETERMINATION OF FATTY ACID PROFILE

[Page 1, clause 5.1.2.1, (c) (5)] — Delete.

[Page 1, clause 5.1.2.1, (c)] — Insert the following at the end:

‘(d) Capillary column — fused silica capillary 100 m and 0.25 mm that is coated with SP-2560 or CP-Sil 88), 100 percent cyanopropylsilicone stationary phase to a thickness of 0.20 µm. Commercially prepared columns are available from different suppliers.

NOTE — The use of 100 m, 0.25 mm ID, 0.20 µm film thickness columns with SP-2560 or CP-Sil 88 as the stationary phase are recommended, as the separation capacity of these columns is sufficient to resolve most C18 : 1 trans- and cis-isomers. If this separation is not required, a 50 m or 60 m column can also be used. However, some 50 m or 60 m long columns might also achieve this separation mostly for vegetable oils. Other types of columns (BPX70, DB-23, HP-88, Rtx-2330, SP-2330, SP-2380, etc) are also possible, but a shift in the elution order is possible. For fast GC analysis, short columns are also possible (10 m to 15 m), but with limited information which in certain cases, will not be a problem.

(e) Detector — Flame ionization detector – capable of being heated to a temperature above that of the column.’

(Page 2, clause 5.1.3.1) — Insert the following at the end:

‘H₂ carrier gas with 99.9 percent purity can also be considered for better resolution’.

(Page 2, clause 5.2.2) — Substitute the following for the existing:

‘5.2.2 Reagents and Apparatus’

(Page 2, clause 5.2.2.5) — Insert the following at the end:

‘5.2.2.6 Carrier gas — see 5.1.3.1.

5.2.2.7 Auxillary gas — see 5.1.3.2.

5.2.2.8 Reference standards — see 5.1.3.3.

5.2.2.9 Apparatus — see 5.1.2.’

(Page 3, clause 5.3, first sentence) — Substitute the following for the existing:

‘For the packed column, maintain GC column with isocratic condition temperature of 185 °C and a detector temperature of 200 °C. For the capillary column, a temperature ramp from 120 °C to 240 °C at a rate of 4 °C/min, followed by holding at 240 °C for 7 min, is recommended for better separation of cis- and trans-isomers of C18 : 1. The inlet and detector temperatures should be maintained at 250 °C.